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EDUCATION

Grambling State University

Grambling, LA

Bachelor of Science in Computer Science — GPA: 3.93

Expected May 2028

- **Awards:** Dean's List (2024, 2025), Presidential Scholarship, Vanda African Math Olympiad '18 Gold Medalist
- **Coursework:** OOP, Software Engineering, Data Structures & Algorithms, Operating Systems, Computer Networks, Cloud Computing

EXPERIENCE

Develop for Good

Remote

Engineer

May 2026 – Aug 2026

- Selected from a competitive applicant pool to engineer a full website redesign for LiberArte Inc., a NYC nonprofit advancing racial, social, and climate justice through the arts.
- Leading visual design, information architecture, and low/no-code implementation to deliver a high-performing, accessible, mobile-responsive platform supporting storytelling, community engagement, and fundraising.
- Collaborating in weekly client-facing meetings and cross-functional team syncs; managing project scope and delivery milestones across a 16-week engagement (5–10 hrs/week).

Outamation

Remote

Extern

May 2026 – Present

- Built modular, AI-powered pipelines to process 200+ page mortgage blob files — combining OCR (Tesseract, PaddleOCR), PDF parsing (PyMuPDF), and RAG techniques for intelligent data extraction, classification, and search.
- Developed a document retrieval system using LlamaIndex and Retrieval-Augmented Generation (RAG), optimized for multi-document mortgage blobs; enhanced precision through chunk tuning, metadata filtering, and evaluation of open-source LLMs including Mistral and Phi-2.
- Conducted end-to-end evaluation of the document intelligence system on 200+ page mortgage blobs — benchmarking OCR accuracy, RAG retrieval quality, and routing performance; delivered a technical report outlining model trade-offs, optimization strategies, and final deployment recommendations; built a demo UI.

Grambling State University

Grambling, LA

Undergraduate Research Assistant

Jan 2025 – Present

- Researching ML-based adaptive OTP verification; engineered an OTP workflow using AES-GCM encryption and Redis for secure token storage, single-use enforcement via TTL-based expiration, and replay attack mitigation.
- Analyzed behavioral anomaly signals in authentication logs to identify suspicious access patterns and proposed risk-based controls to strengthen system security posture.
- Documented system architecture and experimental findings; contributing research outputs to support ongoing lab publications and security infrastructure improvements.

Hubtel Ghana

Accra, Ghana

Software Engineer Intern

May 2023 – Aug 2023

- Built an IoT data pipeline (Kafka Connect, MQTT, PostgreSQL) for real-time sensor processing, improving infrastructure downtime detection by 40%.
- Wrote automated unit and integration tests (Spring Boot, JUnit, Postman) increasing reliability and edge-case coverage across distributed services.
- Shipped Firebase push notifications delivering real-time device and workflow status updates to end users with sub-10ms latency.

PROGRAMS & FELLOWSHIPS

AI4ALL

Ignite Fellow

Remote

Mar 2026 – Present

- Competitively selected for AI4ALL's Ignite Program — a 13-week AI/ML portfolio project track focused on building responsible, socially impactful AI systems with mentorship from industry AI professionals.
- Developing an AI/ML portfolio project applying machine learning to a real-world problem, with emphasis on fairness, interpretability, and ethical deployment.
- Participating in a 7-week career readiness track including resume workshops, technical interview preparation, and direct exposure to AI professionals across the industry.

Nvidia Summer Bridge Program

Participant

Virtual

Summer 2025

- Competitively selected for a 6-week program offering structured exposure to NVIDIA's AI and GPU ecosystem, engineering culture, and early-career development pathways.
- Paired with an NVIDIA engineer mentor for weekly 1:1s covering GPU architecture, parallel computing fundamentals, and navigating a career in systems-level software.
- Attended technical sessions on CUDA programming and accelerated computing; applied concepts in a self-directed exploration of GPU memory optimization patterns.

Basta x Google — Code2Career Program (G-SWEP)

Software Engineering Fellow

Virtual

Aug 2024 – Dec 2024

- Competitively selected for a 10-week program pairing first-generation CS students of color with Google Software Engineers for 1:1 technical mentorship and career guidance.
- Completed weekly live workshops on Data Structures & Algorithms and technical interview preparation; practiced problem-solving patterns under timed conditions with mentor feedback.
- Reviewed system design concepts and mock interview performance with mentor; applied learnings to strengthen internship application strategy across competitive recruiting cycles.

ALX Africa

Software Engineering Program

Remote

Apr 2024

- Completed ALX Africa's rigorous, project-based software engineering curriculum covering full-stack development, systems programming, and professional software engineering practices.
- Built and shipped multiple projects under real-world constraints and strict deadlines, developing strong habits around version control, code review, and iterative delivery.
- Gained foundational depth in Linux, shell scripting, C programming, and web development as part of a globally recognized African tech talent pipeline.

VIRTUAL EXPERIENCE PROGRAMS

JPMorgan Chase — Advanced Software Engineering Simulation

Participant | Forge

Remote

Jan 2026 – Feb 2026

- Integrated Kafka into a Spring Boot microservice to consume and deserialize high-volume transaction messages using a configurable topic and embedded Kafka test framework; built Midas Core, a real-time financial transaction processing system.
- Implemented transaction validation and persistence with Spring Data JPA and H2 SQL, including entity modeling and balance updates; connected service to an external REST Incentive API via RestTemplate.
- Developed a REST endpoint for user balance queries with clean architectural separation; verified system behavior end-to-end using Maven test suites and debugger-driven inspection.

Wells Fargo — Software Engineering Simulation

Participant | Forge

Remote

Jan 2026 – Feb 2026

- Analyzed requirements for a financial portfolio management system and determined the data model needed to support advisor-client-portfolio relationships at scale.
- Designed a relational entity model with JPA entities (FinancialAdvisor, Client, Portfolio, Security) and produced a full ERD capturing cardinality and foreign key constraints.
- Implemented the data model in Spring Boot using IntelliJ, published the codebase to GitHub, and validated entity relationships through end-to-end testing.

Hewlett Packard Enterprise — Software Engineering Simulation

Participant | Forge

Remote

Feb 2026

- Wrote a detailed proposal for a RESTful web service to manage an employee directory, defining endpoints, request/response contracts, and error-handling behavior.
- Built the proposed web server in Java Spring Boot, implementing handlers for HTTP GET and POST requests and support for JSON data upload via REST endpoints.
- Developed and executed a comprehensive unit test suite to assess application performance and validate correctness across edge cases and expected request flows.

Quantium — Data Analytics Simulation

Participant | Forage

Remote

Feb 2026

- Performed data preparation and customer analytics on transaction datasets, cleaning and transforming raw data to extract actionable commercial insights for a retail business unit.
- Identified benchmark stores for conducting uplift testing on trial store layouts, applying statistical methods to enable evidence-based evaluation of layout changes.
- Synthesized analytical findings into a comprehensive report for the Category Manager, delivering data-driven commercial recommendations that directly informed strategic decisions.

PROJECTS

Spring Boot Actuator Dashboard | *Java, Spring Boot, SSE, CycloneDX, Maven* github.com/emmaeng700/javaprojects

- Built a full production observability stack featuring live actuator endpoint monitoring, Spring Cache integration (@Cacheable/@CacheEvict across 3 named caches), and SBOM generation via CycloneDX for supply chain transparency.
- Implemented an ActuatorProxyController bridging the management port (9001) to the app port (9000), solving CORS constraints by proxying all actuator requests through dedicated REST routes via RestTemplate.
- Delivered a dark mode UI with real-time SSE event streaming, search filtering, keyboard shortcuts, color-coded status indicators, and templated endpoint forms with parameter validation.

MedStroke AI | *Python, FastAPI, TensorFlow, React*

github.com/emmaeng700/stroke-diagnoser

- Built a full-stack stroke triage system with NIHSS scoring logic and automated tPA eligibility determination using a Random Forest risk model trained on patient blood-sample features.
- Designed a FastAPI backend serving ML inference results with structured JSON responses; integrated TensorFlow model loading and prediction pipeline with input validation.
- Developed a clinician-facing React interface that renders stroke-risk recommendations, eligibility outcomes, and feature attribution in a clean, actionable dashboard.

Go Container (Minimal Container Runtime) | *Go, Linux*

github.com/emmaeng700/distributed-systems

- Built a minimal container runtime in Go using Linux namespaces (UTS, PID, MNT) and cgroups for process isolation and resource limits, mirroring core Docker internals.
- Implemented a chroot filesystem sandbox with mounted pseudo-file systems (/proc, /sys) to provide isolated process environments with controlled system access.
- Validated isolation behavior with documented architecture and behavioral tests; produced a technical write-up mapping implementation decisions to real container runtime design patterns.

NanoSQL (SQLite-like Database Engine) | *C*

github.com/emmaeng700/C-Database

- Engineered a SQLite-like relational database engine in C with a B-tree indexing structure for efficient key lookups and persistent page-based storage across disk-backed files.
- Implemented a CLI supporting INSERT and SELECT operations with a custom tokenizer and statement parser; designed the storage layer around fixed-size pages for predictable I/O.
- Built node splitting and rebalancing logic to maintain B-tree invariants under high-load write operations, reducing write corruption risk and preserving query correctness.

LEADERSHIP & ACTIVITIES

National Society of Black Engineers (NSBE)

Chapter Treasurer

Grambling State University

Jan 2025 – Present

- Managed chapter budget and financial reporting; secured \$3K in corporate partnerships to fund technical events, workshops, and mentorship programs for a 300+ member community.
- Coordinated with corporate sponsors and university administration to allocate funding toward high-impact programming aligned with NSBE's mission to increase Black representation in engineering.

- Maintained transparent financial records and presented budget updates at chapter meetings, building trust and accountability across the chapter leadership team.

Open-Source Contributor

Remote

emmanuelopponga.com/opensourcecontributions

Feb 2026 – Present

- Merged PRs to pgjdbc, kubernetes-client/java, and Apache Kafka — contributing to production codebases powering infrastructure at Meta, LinkedIn, JPMorgan, and thousands of enterprise engineering teams worldwide.
- Actively investigating a StrictMode regression in facebook/react; signed 4+ CLAs and submitted 20+ PRs across major open-source projects spanning database drivers, container orchestration, and streaming systems.
- Maintains a live open-source contribution tracker on personal portfolio site, reflecting real-time PR status across all active repositories.

JPMorgan Chase — Data for Good Hackathon

Plano, TX

Selected Participant (Competitive Invite)

Apr 2026

- Competitively selected via invite to collaborate on a social-impact data/ML project in a fast-paced, team-based hackathon environment hosted by JPMorgan Chase.
- Applied data analysis and machine learning techniques to a real-world community problem, translating raw datasets into actionable insights and a working prototype under time pressure.
- Collaborated cross-functionally with teammates from diverse backgrounds; presented findings and technical approach to hackathon judges and JPMorgan mentors.

The Lantern Network

Washington, D.C.

Student Ambassador

Dec 2025 – Present

- Selected as a mentee and student ambassador; attended the 13th Annual CLRM Conference and 4th Annual EFMR Conference in Washington, D.C. as an all-expenses-paid invited participant.
- Networked with construction finance professionals, environmental engineers, and EHS industry leaders across two premier invitation-only industry forums.
- Represented Grambling State University in professional industry settings, building relationships with senior practitioners and expanding professional network across environmental and finance sectors.

ColorStack

Remote

Member

Aug 2024 – Present

- Active member of ColorStack, a community dedicated to increasing the success of Black and Latinx CS students in pursuing and thriving in technical careers.
- Participates in community events, job referral networks, and peer knowledge-sharing channels focused on internship recruiting, technical interview prep, and career development.
- Leverages ColorStack's network to stay current on recruiting timelines, company-specific processes, and industry best practices shared by peers and alumni across top tech companies.

Karat

Remote

Member

Oct 2024 – Present

- Member of Karat's engineering interview community, gaining access to structured technical interview practice and assessment resources used by top companies globally.
- Engaged with Karat's interview readiness content to sharpen problem-solving under real interview conditions, including timed coding challenges and behavioral question frameworks.
- Applies Karat's interview frameworks and feedback to continuously refine technical communication, code clarity, and problem decomposition skills ahead of recruiting cycles.

CERTIFICATIONS

HackerRank

Remote

Software Engineer Certification | Software Engineer Intern Certification

Dec 2025

- Passed both HackerRank certification exams, demonstrating proficiency in algorithmic problem solving, SQL query design, and REST API development.
- Software Engineer certification assessed advanced skills in MySQL, REST APIs, and data structure manipulation; Intern certification focused on foundational problem solving and SQL.
- Certifications verified by HackerRank's standardized assessment platform, providing third-party validation of core engineering competencies for recruiting pipelines.

TECHNICAL SKILLS

Languages: Python, Java, Go, C, C++, TypeScript/JavaScript, SQL

Frameworks & Libraries: Spring Boot, React, Next.js, FastAPI, TensorFlow, JUnit, Spring MVC, Spring Data JPA

Infrastructure & Cloud: Kafka, Redis, PostgreSQL, Docker, Kubernetes, Linux, AWS (S3, DynamoDB, Bedrock), Firebase, H2

Tools: Git/GitHub, Bash, CI/CD, Figma, Postman, Maven, CycloneDX, Spring Actuator, IntelliJ

Concepts: Distributed Systems, Microservices, REST APIs, Event-Driven Architecture, IoT Pipelines, System Design, OOP, Data Structures & Algorithms, Data Analytics, GPU/Parallel Computing (CUDA)